Explainable Artificial Intelligence (XAI) Mini Project (30 Marks)
Report Preparation Guidelines

Report Preparation

The report should be 40 to 45 pages content covers the given sections with following formatting guidelines.

Parameter	Requirement
Paper size	A4
Margins	1 inch on all four sides
Font	Times New Roman
Paper title	16 pt, bold, centred
Author names and affiliations	11–12 pt, centred
Abstract and keywords	10–11 pt
Main section headings	12 pt, bold
Subsection headings	11 pt, bold
Body text	11 pt, justified
Line spacing	1.15 used consistently
Figures and tables	Numbered consecutively, with captions
Figure captions	10 pt, centred below the figure
Table captions	10 pt, placed above the table
Equations	Centred and numbered when necessary
Page numbering	Bottom centre or bottom right
References	10 pt, IEEE style

Suggested Sections:

1. **Abstract:** Write a concise summary of the entire project in approximately 150–250 words (Not exceeding 250 words)
2. **Keywords:** Provide 4–6 keywords below the abstract.
3. **Introduction:** Explain the background and motivation of the project. Describe the Limitations of conventional ML/DL models, particularly their lack of interpretability where relevant. Define research objectives and expected contributions.
4. **Related Work:** Review existing research related to the selected problem and the XAI methods. Atleaset 50 reference papers from reputed journals published in the year (2020 to 2026). Avoid conference papers, book chapters, white papers etc. All reviewed papers must be cited in the report and listed in the references.Suggested comparison table:

Author and year	Model / application	XAI technique	Key finding / limitation
Paper 1	Existing ML/DL approach	SHAP	Summarise the finding

5. **Dataset Description and Pre-processing:** Describe the dataset and how it was prepared for experimentation. Include the Dataset name, source, and URL. Students must clearly identify whether the dataset is publicly available, collected by them, or obtained from another source. Explain the pre processing strategy with neat representation.
6. **Proposed Methodology:** Explain the complete workflow of the proposed system. Overall system architecture or workflow diagram is must. Define the model architecture and important hyperparameters. Integration of the XAI technique into the trained model. A step-by-step algorithm or pseudocode describing the workflow.
7. **Experimentation and Result analysis:** Describe how the proposed system was implemented. Specify the programming language and development environment. Specify the model configuration and training parameters. In result analysis present the actual experimental outcomes of the implemented system. Explain the training and validation results, test set performance, appropriate evaluation metrics, Confusion matrix, Graphs and tables presenting the results and brief interpretation of the observed performance.
8. **Explainable AI Integration and Analysis:** This is the most important section of the report. Students must demonstrate how the XAI technique explains the behaviour of their trained model, rather than merely describing the technique theoretically. Student should explain the name and justification of selected XAI method, explanation is global, local, or both. Explain how the technique was applied to the trained model.
9. **Ablation Study:** Conduct an ablation study to investigate how individual components of the proposed system affect its performance and explainability. Students should compare the complete model with selected variants in which one component is removed, replaced, or modified while keeping other experimental conditions as consistent as possible.
10. **Conclusion and Future Work:** Summarize the project and its main contributions.
11. **References:** Include all 50 research papers cited in the report, dataset publications, official software documentation, and relevant technical resources. Use a consistent IEEE reference style.

    Example format:

    [1] A. Author and B. Author, “Title of the research paper,” Journal or Conference Name, vol. X, no. Y, pp. XX–XX, year.